Final Report: Newton–Schulz Schedule Design in Muon for Transformer Pretraining

Yicheng Zhao, Sihua Zhang, Ruilin Gao
COE202 Mathematical Foundations of AI, Spring 2026

1 Problem Setup

Problem statement. Muon is a recently proposed optimizer that updates matrix parameters by orthogonalizing a momentum-like update direction through a short Newton–Schulz (NS) iteration [2, 3]. Recent theory has also started to analyze the convergence properties of Muon-style NS updates [4]. In practice, the iteration is controlled by quintic coefficients, so schedule design is not merely an implementation detail. This project asks: *under a fixed transformer pretraining stack, how do different NS coefficient schedules change the spectral geometry of Muon updates, and how do these geometric changes affect validation loss and training cost?*

Mathematical formulation. AdamW treats each entry of a weight matrix with coordinatewise first- and second-moment statistics. Muon instead treats a hidden-layer weight as a matrix and attempts to make the update direction more semi-orthogonal. For a rectangular matrix X, the target geometry is

$$X^\top X \approx I_n \quad \text{if } m \geq n, \qquad XX^\top \approx I_m \quad \text{if } m < n. \tag{1}$$

Equivalently, if $X = U\Sigma V^\top$, the ideal polar factor $UV^\top$ has all nonzero singular values equal to one. Muon approximates this polar factor by repeatedly applying a quintic matrix polynomial.

The high-level Muon update for a matrix parameter is summarized in Algorithm 1. Non-matrix parameters such as embeddings and normalization parameters are still updated by Adam-style rules, so the defining difference from AdamW lies in the matrix update path.

The NS step acts on singular values. If X_k is the matrix after the k-th iteration, then

$$X_{k+1} = a_k X_k + b_k X_k X_k^\top X_k + c_k (X_k X_k^\top)^2 X_k, \tag{2}$$

or the aspect-ratio-swapped equivalent for tall matrices. At the singular-value level,

$$\sigma_{k+1} = p_k(\sigma_k), \qquad p_k(\sigma) = a_k\sigma + b_k\sigma^3 + c_k\sigma^5. \tag{3}$$

Thus a schedule $\{(a_k, b_k, c_k)\}_{k=1}^T$ defines a composed spectral map p_T that controls how aggressively singular values are moved toward one. In this sense, Muon is a family of optimizers parameterized by spectral maps.

We study AdamW and four Muon schedule families:

- **stable5**: five stable NS steps $(2.0, -1.5, 0.5)$;
- **fast5**: five aggressive Muon steps $(3.4445, -4.7750, 2.0315)$;
- **manual**: a fast-to-stable schedule, e.g. `manual_T5_f3_s2` means three fast steps followed by two stable steps;
- **Polar Express**: quintic coefficients generated from matrix-sign theory and a lower-bound assumption [1].

2 Technical Approach

Controlled experimental design. The central goal is to isolate the effect of the coefficient schedule. We therefore fix the model, data source, token budget, batch structure, learning-rate schedule, and all non-schedule optimizer hyperparameters. AdamW is included as an external baseline because it does not use NS orthogonalization. Within the Muon family, the primary treatment variable is the schedule-defined spectral map.

All experiments are run on one RTX 4090 GPU in BF16 precision using FineWeb-10B tokenized shards [5]. The model is an 11-layer pre-norm Transformer with model dimension 768, 6 attention heads, head dimension 128, MLP ratio 4, and sequence length 2048. The training budget is fixed at 100M tokens, with 131,072 tokens per optimizer step, 16 gradient-accumulation steps, 2% warmup, and cosine decay to 20% of the peak learning rate.

The optimizer settings are fixed as follows:

- AdamW baseline: learning rate 6×10^{-4}, $(\beta_1, \beta_2) = (0.9, 0.95)$, weight decay 0.1.
- Muon family: matrix learning rate 0.023, momentum 0.95, $\beta_2 = 0.9$, weight decay 1.2.

Algorithm 1 High-level Muon update for a matrix parameter

Initialize momentum state M_t
for $t = 1, 2, \dots$ **do**
    Compute matrix gradient $G_t = \nabla_W \mathcal{L}_t(W_{t-1})$
    Update momentum direction $M_t \leftarrow \mu M_{t-1} + G_t$
    Obtain a semi-orthogonalized direction O_t by applying NS iteration to M_t
    Update parameters: $W_t \leftarrow W_{t-1} - \eta_t O_t$
end for

Table 1: Core $T = 5$ training metrics under the fixed 100M-token budget.

Method	Schedule	Final loss	AUC over tokens
AdamW	`adamw`	5.3737	5.9677
Stable NS	`stable5`	4.5273	5.3166
Manual	`manual_T5_f3_s2`	4.3176	5.0295
Fast Muon	`fast5`	**4.2828**	4.9340
Polar Express	`pe_T5_l3e-3`	4.2873	4.9462
Polar Express	`pe_T5_l1e-3`	4.2902	**4.9338**

Implemented Muon variant and diagnostics. The implementation is a normalized Muon variant with a Nesterov-style lookahead path. Let G_t denote the matrix gradient and B_t the momentum buffer:

$$B_t = mB_{t-1} + (1 - m)G_t, \tag{4}$$

$$g_{\mathrm{pre},t} = mB_t + (1 - m)G_t, \tag{5}$$

$$g_{\mathrm{post},t} = \mathcal{P}(g_{\mathrm{pre},t}), \tag{6}$$

where $\mathcal{P}$ is the schedule-defined NS map. The measured geometry is therefore the geometry of the real implemented update, not an idealized textbook Muon direction.

The study uses three run modes. `train` records validation-loss trajectories, `benchmark` measures end-to-end wall-clock time with synchronization, and `spectral` samples real update matrices for SVD diagnostics. For a sampled matrix X, we use the short-side semi-orthogonality error

$$\varepsilon(X) = \begin{cases} \|X^\top X - I\|_F, & X \text{ tall or square,} \\ \|XX^\top - I\|_F, & X \text{ wide.} \end{cases} \tag{7}$$

This metric is recorded for `buffer_post`, `g_pre`, and `g_post`. We also compute validation-loss AUC over tokens and over wall-clock time. These AUCs summarize the average validation loss along the whole curve, so they are more stable than a local loss-decrease slope over an arbitrary interval.

3 Main Results

3.1 Fixed-T Coefficient Schedules

The fixed five-step comparison shows that coefficient choice is already enough to change training behavior. Table 1 and Figure 1 compare the core $T = 5$ schedules. AdamW is clearly separated from all Muon variants. Within the Muon family, `stable5` is conservative and substantially worse, `manual_T5_f3_s2` lies in the middle, and `fast5` together with the better Polar Express schedules forms the leading group.

The result also explains why the original fast Muon coefficients should not be treated as arbitrary tuning constants. The stable NS map is mathematically conservative, but its conservative singular-value amplification is not ideal for early pretraining. The fast coefficients appear to provide a useful amount of aggressive amplification while remaining stable enough for this model and token budget.

3.2 Follow-up Checks: Seeds, Depth, and Lower Bounds

The multi-seed check makes the main training conclusion conservative rather than stronger. For three representative schedules, we repeated training with seeds 0, 1, and 2. The mean results in Table 2 show that `fast5` remains the best practical baseline, but the gaps are small. The standard deviations are of the same order as the differences among the top schedules, so the correct interpretation is not that one schedule decisively dominates, but that more complex geometry does not reliably improve final validation loss in the current 100M-token regime.

Increasing manual depth helps at first but is not monotone. The manual family improves sharply from `manual_T5_f3_s2` to the $T = 7, 8, 9, 10$ range. For example, final loss drops from 4.3176 at $T = 5$ to 4.2873 at

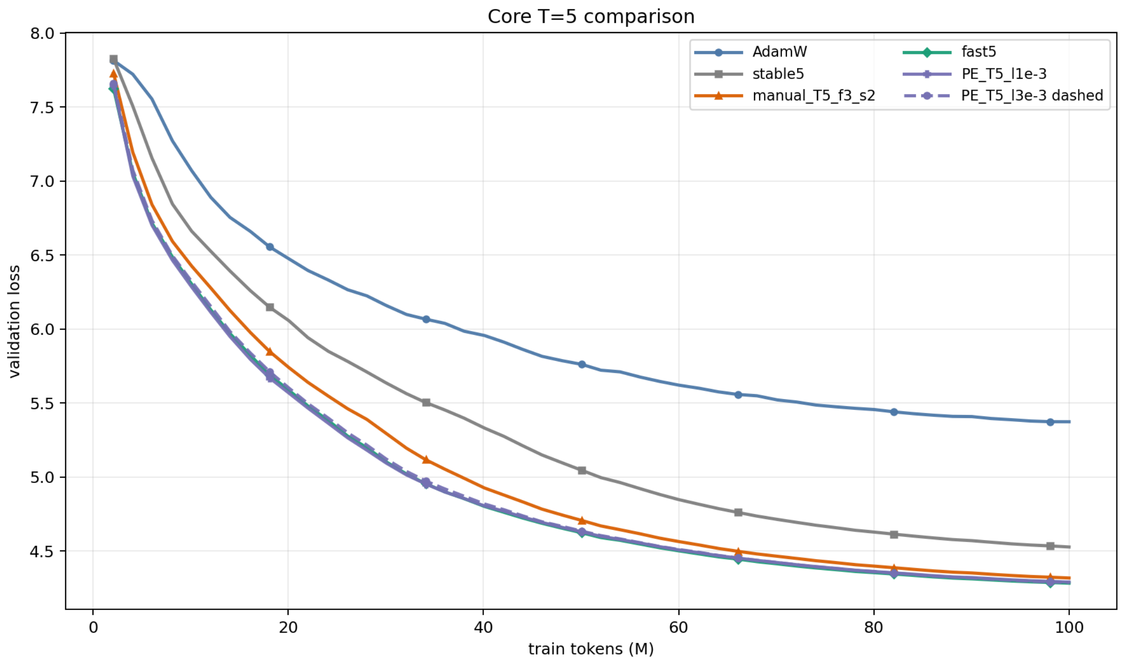

Figure 1: Core $T = 5$ validation-loss curves. All curves use the same model, data, token budget, batch setting, and learning-rate schedule. Lower curves indicate better token efficiency.

Table 2: Multi-seed final validation loss for representative schedules.

Schedule	Mean final loss	Std. over seeds
`fast5`	**4.2892**	0.0056
`manual_T9_f4_s5`	4.2962	0.0038
`pe_T9_l3e-5`	4.2998	0.0044

`manual_T7_f4_s3`. After that, the results lie in a narrow band: `manual_T8_f5_s3`, `manual_T9_f3_s6`, `manual_T9_f4_s5`, and `manual_T10_f5_s5` have final losses between 4.2887 and 4.2949. Thus, depth is useful as a way to approach the leading group, but more NS iterations are not automatically better.

Polar Express lower bound is a real algorithmic parameter. At fixed $T = 5$, changing the lower bound changes both final loss and AUC: `pe_T5_l3e-3` reaches 4.2873, `pe_T5_l1e-3` reaches 4.2902, `pe_T5_l3e-4` reaches 4.2998, and `pe_T5_l3e-5` reaches 4.3125. This is not merely a cosmetic hyperparameter; it changes the singular-value interval assumed by the Polar Express construction and therefore changes the composed spectral map. Increasing PE depth to $T = 9$ or $T = 10$ improves the low-bound setting, but it still does not give a robust loss advantage over `fast5`.

3.3 Cost and Spectral Geometry

Coefficient changes at fixed depth have negligible wall-clock effect. At $T = 5$, the Muon schedules all finish in about 1095–1099 seconds, while AdamW finishes in 1039.73 seconds. The AdamW baseline is faster because it does not execute the NS orthogonalization path. In contrast, changing scalar coefficients inside a fixed five-step Muon schedule barely changes runtime. Deeper $T = 9$ schedules are naturally more expensive: `manual_T9_f4_s5` takes 1123.53 seconds and `pe_T9_l3e-5` takes 1119.42 seconds. This motivates reporting both token efficiency and wall-clock AUC instead of ranking schedules only by final loss.

Geometry changes are much larger than loss changes. The strongest evidence for the mathematical content of the project comes from spectral diagnostics. Table 3 reports the post-orthogonalization error for representative schedules. Deeper manual and PE schedules make `g_post` much closer to semi-orthogonal than `fast5`, especially in attention matrices. However, these geometric improvements do not translate into lower multi-seed validation loss. This is the central tension in the results: better semi-orthogonality is real and measurable, but its relationship to language-model validation loss is not monotone at this training scale.

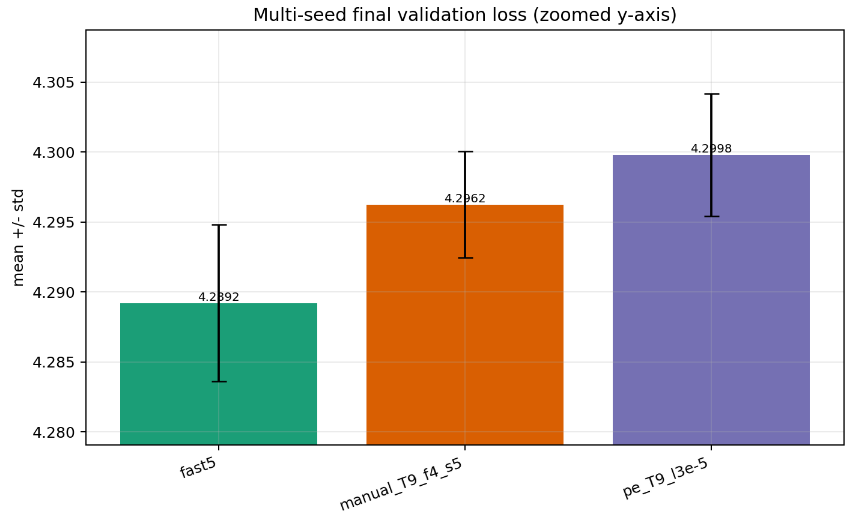

Figure 2: Multi-seed final validation loss over seeds 0/1/2. The leading schedules are close, so the correct conclusion is that `fast5` is the most reliable practical baseline in this setting rather than a decisive universal winner.

Table 3: Representative spectral diagnostics. Lower values mean the update is closer to semi-orthogonal.

Schedule	Overall `g_post`	Attention	MLP
`stable5`	0.9006	–	–
`fast5`	0.6444	0.5146	0.7743
`manual_T5_f3_s2`	0.6866	–	–
`manual_T9_f4_s5`	0.4431	0.1776	0.7086
`pe_T5_l1e-3`	0.5452	–	–
`pe_T9_l3e-5`	**0.3781**	**0.0481**	**0.7081**

4 Singular-Value Map Interpretation

The empirical results are easier to interpret through the scalar map $p_T(\sigma)$. A useful diagnostic is the gain

$$\text{gain}(\sigma) = \frac{p_T(\sigma)}{\sigma}, \tag{8}$$

which measures how strongly a small singular value is amplified by the full schedule. At $\sigma = 10^{-5}$, the gain values are approximately 32 for `stable5`, 485 for `fast5`, 4502 for `manual_T9_f4_s5`, and 71309 for `pe_T9_l3e-5`. This ordering explains the spectral results: more aggressive schedules can reduce `g_post` error much more strongly.

The same interpretation also explains why the final training conclusion must be modest. A schedule that aggressively pushes singular values toward one may create a cleaner semi-orthogonal update, but pretraining loss depends on the interaction among update direction, momentum trajectory, stochastic gradients, and model architecture. In our results, attention projections show the largest geometric response to stronger schedules, while MLP matrices respond less. This suggests that schedule design should be understood as a structured way of changing optimization geometry, not as a one-dimensional knob where "more orthogonal" always means "better loss."

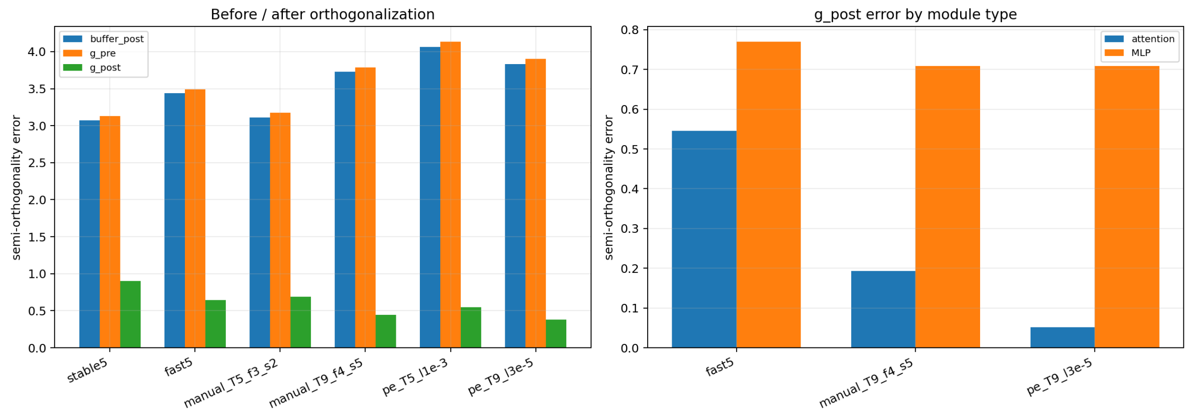

Figure 3: Spectral diagnostics. Left: `buffer_post`, `g_pre`, and `g_post` errors show how much the NS map changes the update. Right: attention matrices are more schedule-sensitive than MLP matrices.

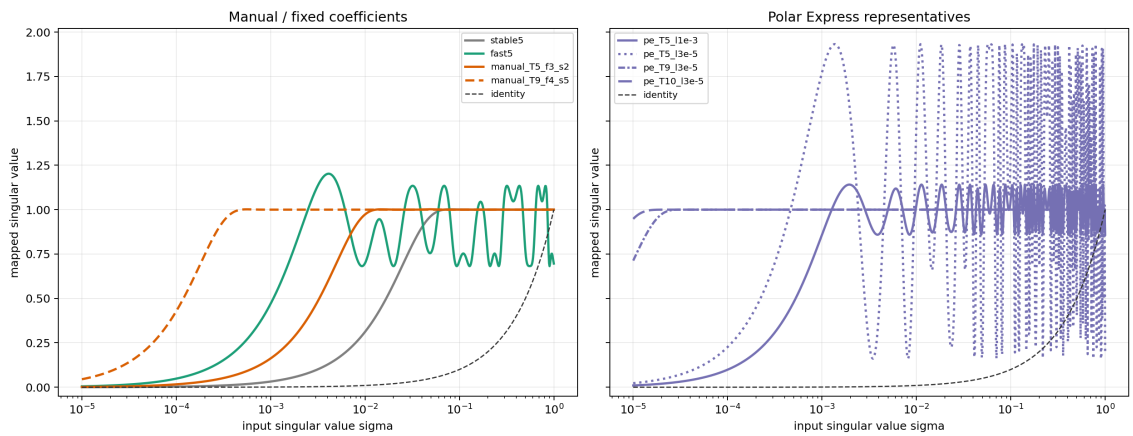

Figure 4: Composed singular-value maps. Stable schedules are smooth and conservative, while deeper manual and PE schedules amplify small singular values more aggressively. The stronger maps improve geometric semi-orthogonality but do not guarantee lower validation loss.

5 Conclusion

This project studied Muon as a family of Newton–Schulz spectral transformations rather than as a single optimizer. The controlled $T = 5$ comparison shows that coefficient sequences matter: `stable5` is too conservative, `manual_T5_f3_s2` is intermediate, and `fast5` together with well-chosen Polar Express schedules forms the leading group. The multi-seed follow-up makes the strongest practical statement clear: in the current 100M-token setting, `fast5` is the most reliable baseline, while deeper manual and PE schedules do not robustly lower final validation loss.

At the same time, the spectral diagnostics show that the more complex schedules are not meaningless. They substantially reduce post-orthogonalization error, especially in attention matrices, and their behavior can be explained by the composed singular-value map $p_T(\sigma)$. The main scientific conclusion is therefore not that a new schedule simply wins the benchmark. Rather, different NS schedules create controlled changes in update geometry, and those changes only partially translate into validation-loss improvements. This gives a more nuanced answer to the mathematical question behind Muon: semi-orthogonalization is useful, but the optimal amount and shape of spectral correction depend on the training dynamics.

The study remains limited to one model scale, one 100M-token budget, and a small multi-seed set for representative schedules. Future work should repeat the strongest schedules at larger token budgets and analyze whether the attention-dominant spectral differences persist at scale. A complementary theoretical direction is to connect the fixed points, contraction regions, and lower-bound assumptions of the composed quintic map more directly to the measured training trajectories.

References

[1] N. Amsel, D. Persson, C. Musco, and R. M. Gower. The polar express: Optimal matrix sign methods and their application to the muon algorithm, 2025. Version 5 revised in 2026.

[2] K. Jordan. Muon: An optimizer for hidden layers in neural networks. Blog post, 2024. URL `https://kellerjordan.github.io/posts/muon/`.

[3] K. Jordan, J. Bernstein, B. Rappazzo, et al. modded-nanogpt: Speedrunning the nanogpt baseline. GitHub repository, 2024. URL `https://github.com/KellerJordan/modded-nanogpt`.

[4] G. Y. Kim and M.-h. Oh. Convergence of muon with newton–schulz. In *International Conference on Learning Representations*, 2026.

[5] G. Penedo, H. Kydlíček, L. Ben Allal, A. Lozhkov, M. Mitchell, C. Raffel, L. von Werra, and T. Wolf. The fineweb datasets: Decanting the web for the finest text data at scale. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2024.